

Bansilal Ramnath Agarwal Charitable Trust's

Vishwakarma Institute of Technology, Pune-37

(An autonomous Institute of Savitribai Phule Pune University)



Artificial Intelligence and Data Science Department

MLOps Programming

Division	D
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PART A

```
PS C:\VITP Academics\Sem 5\MLOps> python -m pip show dvc
Name: dvc
Version: 3.67.1
Summary: Git for data scientists - manage your code and data together
Home-page:
Author:
Author-email: Dmitry Petrov <dmitry@dvc.org>
License-Expression: Apache-2.0
Location: C:\Users\Pruthviraj Mohite\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_
Requires: attrs, celery, colorama, configobj, distro, dpath, dulwich, dvc-data, dvc-http, dvc-obj
funcy, grandalf, gto, hydra-core, iterative-telemetry, kombu, networkx, omegaconf, packaging, pat
ml, scmrepo, shortuuid, shtab, tabulate, tomlkit, tqdm, voluptuous, zc.lockfile
Required-by:
PS C:\VITP Academics\Sem 5\MLOps> python -m dvc --version
3.67.1
```

PART B

```
(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> dvc init
Initialized DVC repository.

You can now commit the changes to git.

+-----+
|               |
| DVC has enabled anonymous aggregate usage analytics. |
| Read the analytics documentation (and how to opt-out) here: |
| <https://dvc.org/doc/user-guide/analytics> |
|               |
+-----+

What's next?
-----
- Check out the documentation: <https://dvc.org/doc>
- Get help and share ideas: <https://dvc.org/chat>
- Star us on GitHub: <https://github.com/treeverse/dvc>
```

PART C

```
student.csv X
DatasetVersioning > data > student.csv > data
1 RollNo,Name,Marks
2 1,Amit,80
3 2,Rahul,72
4 3,Priya,91
5 4,Sneha,88
```

PART D

```
(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> dvc add data/student.csv
100% Adding...|

To track the changes with git, run:

    git add 'data\student.csv.dvc' 'data\.gitignore'

To enable auto staging, run:

    dvc config core.autostage true
(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> |
```

PART E

```
(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> git add .
>> git commit -m "Track dataset using DVC"
[main 4fd2f1e] Track dataset using DVC
2 files changed, 6 insertions(+)
create mode 100644 data/.gitignore
create mode 100644 data/student.csv.dvc
(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> dvc add data/student.csv
100% Adding...|

To track the changes with git, run:

    git add 'data\student.csv.dvc'

To enable auto staging, run:

    dvc config core.autostage true
(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> |

(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> git add .
>> git commit -m "Dataset Version 2"
[main a86ad1a] Dataset Version 2
1 file changed, 2 insertions(+), 2 deletions(-)
(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> git log
commit a86ad1a46e786ad5750e68cf666be38227736522 (HEAD -> main)
Author: Pruthvi-22-Mohite <pruthvirajpmohite22@gmail.com>
Date: Mon Jul 13 13:12:13 2026 +0530

Dataset Version 2

commit 4fd2f1e0e13f9b16be5881ab2571c9c8ffa51416
Author: Pruthvi-22-Mohite <pruthvirajpmohite22@gmail.com>
Date: Mon Jul 13 13:10:55 2026 +0530

Track dataset using DVC

commit 87c69a1685a61c8773a5ca4d2615035708f3a5c4
Author: Pruthvi-22-Mohite <pruthvirajpmohite22@gmail.com>
Date: Mon Jul 13 13:09:10 2026 +0530

Initialize DVC
(.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> |
```

PART F

```
• (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> git checkout 4fd2f1e0e13f9b16be5881ab2
Note: switching to '4fd2f1e0e13f9b16be5881ab2571c9c8ffa51416'.
```

You are in 'detached HEAD' state. You can look around, make experimental changes and commit them, and you can discard any commits you make in this state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may do so (now or later) by using `-c` with the switch command. Example:

```
git switch -c <new-branch-name>
```

Or undo this operation with:

```
git switch -
```

Turn off this advice by setting config variable `advice.detachedHead` to false

HEAD is now at 4fd2f1e Track dataset using DVC

PART G

```
• (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> mkdir ..dvc-storage
```

Directory: C:\VITP Academics\Sem 5\MLOps\DatasetVersioning

Mode	LastWriteTime	Length	Name
d-----	13-07-2026 13:17		..dvc-storage

```
• (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> dvc remote add -d localstorage ../dvc-storage
Setting 'localstorage' as a default remote.
```

```
• (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> git add .
```

```
• >> git commit -m "Configure DVC remote"
[main 8baba21] Configure DVC remote
1 file changed, 4 insertions(+)
```

```
○ (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> █
```

```
• (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> dvc push
```

```
Collecting
Pushing
1 file pushed
```

```
• (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> Remove-Item .\data\student.csv
```

```
• (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> dvc pull
```

```
Collecting
Fetching
Building workspace index
Comparing indexes
Applying changes
A      data\student.csv
1 file added
```

```
○ (.venv) PS C:\VITP Academics\Sem 5\MLOps\DatasetVersioning> █
```

PART H

REFLECTION ANSWERS

1. Why is Git alone not suitable for versioning datasets?

Git is mainly used for code. Large datasets make Git slow and take a lot of storage space. DVC is used to manage large datasets separately.

2. What is the purpose of the .dvc file?

The .dvc file stores information about the dataset (like its version and location). Git tracks this small file instead of the actual large dataset.

3. Explain the difference between git add and dvc add.

git add: Adds files to Git for version control.

dvc add: Adds datasets/models to DVC tracking.

Example:

```
dvc add data/student.csv
```

creates student.csv.dvc

which is tracked by Git.

4. What happens when dvc checkout is executed?

dvc checkout restores the correct version of the dataset according to the current Git version.

5. How does DVC improve reproducibility in ML projects?

DVC helps keep the same version of data, models, and code so that ML experiments can be repeated with the same results.